



**CHANAKYA
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PROJECT REPORT

On

Smart Power Saving System

MCA/MSc 2nd Sem

Team Members

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Abstract

Electricity wastage is a common problem in homes, classrooms, offices, and public buildings where lights remain switched ON even when no one is present. The Smart Power Saving System addresses this issue by automatically controlling lights based on room occupancy. The system uses a PIR (Passive Infrared) sensor to detect human movement and body heat. An ESP32 microcontroller processes the sensor data and updates occupancy status on the Blynk IoT platform through Wi-Fi. When motion is detected, the light turns ON automatically, and when the room becomes vacant, the light turns OFF after a specified delay. This helps reduce unnecessary power consumption and promotes energy-efficient automation.

Introduction

Smart home automation plays a significant role in improving energy efficiency and user convenience. Traditional lighting systems require manual operation, which often results in lights being left ON unnecessarily. The proposed Smart Power Saving System utilizes a PIR sensor, ESP32 microcontroller, and Blynk IoT platform to automate room lighting. The PIR sensor detects changes in infrared radiation emitted by the human body and identifies movement within its sensing range. Based on occupancy status, the system automatically controls lighting and updates information on a cloud-based dashboard.

Objectives

- To detect human presence using a PIR sensor.
- To automatically switch lights ON and OFF.
- To reduce unnecessary electricity consumption.
- To provide real-time occupancy monitoring using Blynk IoT.
- To demonstrate a low-cost smart home automation solution.

Hardware Components

- ESP32 Development Board
- PIR Motion Sensor (HW-416)
- LED
- Breadboard
- Jumper Wires
- USB Cable
- Mobile Phone with Blynk App

Software Components

- Arduino IDE 2.3.10
- ESP32 Board Package
- Embedded C/C++
- Blynk IoT Platform
- Blynk Cloud
- Wi-Fi Network

System Architecture

1. PIR sensor continuously monitors the room.
2. Human motion and infrared heat changes are detected.
3. Sensor sends signal to ESP32.
4. ESP32 processes the input.
5. LED/light turns ON when occupancy is detected.
6. Light turns OFF when no motion is detected for a predefined period.
7. Occupancy status is sent to Blynk Cloud.
8. Dashboard displays room status, bulb status, and energy-saving statistics.

Working Principle

The PIR sensor detects changes in infrared radiation caused by human movement. When a person enters the sensing area, the sensor output becomes HIGH. The ESP32 reads this signal and switches the LED ON. Simultaneously, occupancy information is sent to the Blynk dashboard and displayed as "Occupied". When no motion is detected for a configured delay period, the sensor output becomes LOW. The ESP32 then turns the LED OFF and updates the dashboard status to "Unoccupied".

Blynk Dashboard Features

- Room Status Monitoring
- Occupied / Unoccupied Indicator
- Bulb State Indicator
- Master Override Switch
- Energy Saved Graph
- Real-Time IoT Monitoring

Results

The Smart Power Saving System successfully detected human movement using a PIR sensor and automatically controlled room lighting. The system displayed room occupancy status in real time on the Blynk dashboard. Additional dashboard features such as Bulb State Indicator, Master Override Switch, and Energy Saved Graph improved monitoring and usability. Testing demonstrated reliable motion detection, automatic light control, and reduced energy wastage. The project effectively showcased an IoT-based smart home automation solution.

Advantages

- Reduces electricity wastage.
- Fully automatic operation.
- Low-cost implementation.
- Easy installation and maintenance.
- Real-time monitoring through IoT.
- Scalable for smart homes and offices.

Applications

- Smart Homes
- Classrooms
- Offices
- Libraries
- Laboratories
- Conference Rooms
- Public Buildings

Future Enhancements

- Integration with actual AC bulbs using relay modules.
- Mobile notifications for occupancy events.
- Energy consumption analytics.
- Multiple room monitoring.
- Integration with AI-based occupancy prediction.
- Voice assistant support using Google Assistant or Alexa.

Conclusion

The Smart Power Saving System successfully automates room lighting based on occupancy detection. By combining PIR sensing technology, ESP32, and the Blynk IoT platform, the system provides an efficient and practical solution for energy conservation. The project demonstrates how IoT can be used to create intelligent, cost-effective, and environmentally friendly automation systems for modern buildings.